OR Chapter 2 MCQs

1. What is considered the first 'order of business' for an Operations Research (OR) team when starting a new study?

A. Developing a computer-based procedure for deriving solutions.

B. Formulating a mathematical model to represent the constraints.

C. Conducting a retrospective test using historical data.

D. Studying the relevant system and developing a well-defined statement of the problem.

2. In the context of organizational objectives, what is the primary risk of 'suboptimization'?

A. Focusing exclusively on short-term profits instead of long-run profit maximization.

B. The OR team acts in a decision-making capacity rather than an advisory one.

C. Seeking solutions that are best for only one component of an organization rather than the whole.

D. The model becomes too complex to be solved in a reasonable time frame.

3. Which of the following parties, affected by a firm's social responsibilities, is primarily interested in the 'integrity and a reasonable selling price for their goods'?

A. The owners (stockholders)

B. The employees

C. The government

D. The suppliers

4. If an OR model uses the expression , what are the variables called?

A. Objective functions

B. Constraints

C. Parameters

D. Decision variables

5. What is the primary function of 'sensitivity analysis' in the OR modeling process?

A. To ensure that management identifies the 'right' problem from their viewpoint.

B. To translate the mathematical model into a computer-based algorithm.

C. To check if the mathematical expressions are dimensionally consistent.

D. To determine how the derived solution would change if the values assigned to parameters were different.

6. In model development, what does the term 'tractability' refer to?

A. The ability of the model to be solved using available mathematical techniques and computers.

B. The degree to which the model's predictions correlate with real-world outcomes.

C. The process of enriching a simple model into a more elaborate version.

D. The inclusion of all possible factors that might affect the real-world system.

7. Herbert Simon's concept of 'satisficing' suggests that managers often seek a solution that is:

A. Mathematically optimal according to all constraints.

B. Derived solely from intuitively designed heuristic procedures.

C. A 'good enough' guide for action that meets established criteria.

D. Suboptimal for the organization but optimal for a specific department.

8. What is the primary disadvantage of using a 'retrospective test' to validate an OR model?

A. It cannot account for dimensional consistency within the mathematical equations.

B. It uses the same historical data that may have guided the initial formulation of the model.

C. It is only applicable to linear programming models and cannot be used for complex systems.

D. It requires the use of computer software that may contain hidden bugs.

9. In the IBM spare-parts inventory study, why were functional managers appointed to a 'user team' during the model validation phase?

A. To replace the OR team in the final implementation phase.

B. To perform the complex mathematical calculations required for the Optimizer system.

C. To manually verify the billions of dollars in inventory valuations.

D. To act as advisors to the OR team and build a sense of ownership for the new system.

10. According to the implementation phase guidelines, what is required if significant deviations from the original model assumptions occur over time?

A. The implementation should be immediately halted to avoid further losses.

B. A new retrospective test should be performed using only the most recent day of data.

C. The model should be revisited to determine if modifications are needed.

D. The OR team should be held legally responsible for the inaccuracies.

11. According to the source, what is the primary purpose of an OR team during the 'Defining the Problem' phase?

A. To act in an advisory capacity to management.

B. To focus exclusively on the welfare of a single department.

C. To make the final executive decision for the organization.

D. To solve problems exactly as they are initially described by management.

12. Which of the following describes the 'suboptimization' problem in Operations Research?

A. Using a mathematical model that is too simple to solve the real problem.

B. Finding a solution that is best for one component but not optimal for the overall organization.

C. Gathering data that is too 'soft' or imprecise to be used in a model.

D. Failing to reach a 'satisfactory' level of profit for the firm.

13. In the context of profit-making organizations, why might management adopt a goal of 'satisfactory profits' rather than 'long-run profit maximization'?

A. Profit maximization is legally prohibited for most international corporations.

B. Long-run profit maximization ignores the time value of money.

C. Satisfactory profits are easier to calculate than long-run profits.

D. The relationship between other objectives and long-run profit may be too obscure to incorporate into one goal.

14. What are 'decision variables' in a mathematical model?

A. The fixed constants that appear in the constraints of the model.

B. The final results obtained after the implementation phase.

C. The environmental factors that the OR team cannot influence.

D. The quantifiable decisions to be made, represented as variables like .

15. An objective function like is best described as:

A. A description of the physical laws governing the system.

B. A mathematical expression for the appropriate measure of performance.

C. A summary of the 'soft' data gathered during the initial study.

D. A restriction on the values that can be assigned to decision variables.

16. What is the primary role of 'sensitivity analysis' in the modeling process?

A. To calculate the 'overall measure of performance' for multiple objectives.

B. To determine how the solution would change if the assigned values for parameters were different.

C. To identify which members of management are most sensitive to the team's recommendations.

D. To ensure that all mathematical expressions are dimensionally consistent.

17. Which trade-off is constantly considered during the process of 'model enrichment'?

A. The trade-off between short-run and long-run profit maximization.

B. The trade-off between gathering 'soft' data and 'hard' data.

C. The trade-off between the precision and the tractability of the model.

D. The trade-off between using computers and using manual calculations.

18. According to the source, what distinguishes a 'heuristic procedure' from an 'algorithm'?

A. Algorithms focus on 'what-if' analysis, while heuristics focus on data gathering.

B. Heuristics are computerized, while algorithms are performed manually.

C. Algorithms are only used for models that lack a single 'right' answer.

D. Heuristics are intuitively designed procedures that do not guarantee an optimal solution.

19.How did Herbert Simon describe the difference between 'optimizing' and 'satisficing'?

A. Satisficing is seeking a solution that is 'good enough' for the problem at hand.

B. Optimizing is only possible when 'soft' data is completely eliminated.

C. Satisficing is the process of testing a model using historical data.

D. Optimizing is the 'art of the feasible,' while satisficing is the 'science of the ultimate.'

20. What is the primary activity involved in 'postoptimality analysis'?

A. Writing the computer code for the model's algorithms.

B. The final step of implementing the solution into the organization's operations.

C. Addressing 'what-if' questions about how the optimal solution changes under different assumptions.

D. Gathering initial data before the mathematical model is formulated.

21.A parameter is considered 'sensitive' if:

A. It represents a 'soft' estimate based on an educated guess.

B. Its value cannot be changed without changing the optimal solution.

C. It represents a social responsibility objective like 'employee morale'.

D. It is a variable like that management can control directly.

22. What is a 'retrospective test' in the context of model validation?

A. A test to determine if management is satisfied with the final report.

B. Using historical data to see how the model would have performed in the past.

C. Checking if the mathematical units in the equations are consistent.

D. A survey of employees to measure their morale after implementation.

23. Why is it important to include an individual who did not participate in the model formulation in the model review process?

A. To ensure that the model uses only 'hard' data and ignores 'soft' data.

B. Because only outsiders can perform a 'retrospective test'.

C. To take a fresh look at the overall model and identify obvious errors or oversights.

D. To ensure that the model is kept secret from competitors.

24. What is the primary function of a 'decision support system' in OR?

A. A database used exclusively for gathering 'soft' data.

B. A system that automatically executes decisions without human intervention.

C. A legal framework for enforcing social responsibility policies.

D. An interactive computer-based system that helps managers use data and models to support their decision making.

25. Why is 'documentation' considered a crucial part of the culmination of an OR study?

A. To ensure that the work is reproducible, which is part of the researcher's professional ethical code.

B. To satisfy the requirements of the government and tax authorities.

C. To eliminate the need for future sensitivity analysis.

D. To provide a legal shield against potential failures of the model.

26. In the implementation phase, what is the 'initial order of business' for the OR team?

A. To gather 'soft' data from the five parties affected by the business firm.

B. To perform a final 'retrospective test' using the most recent data.

C. To install the computer software on every employee's workstation.

D. To give operating management a careful explanation of the new system and how it relates to operating realities.

27. Which of the following parties is NOT listed in the source as one of the five typically affected by a business firm's social responsibility?

A. Competitors

B. Employees

C. Government

D. Suppliers

28. The Monsanto Corp. example involved a model with over 1,000 decision variables. How was the objective function for this model expressed?

A. As a double summation: .

B. As an inequality: .

C. As a physical law equation like .

D. As a simple linear equation: .

29. True or False: An OR study should always provide an utopian guarantee that the optimal solution for the model is the absolute best possible solution for the real problem.

A. False

B. True

30. What is the critical risk of using a model that is too complex for the available data?

A. It will force the team to use only 'hard' data.

B. It will automatically lead to 'satisficing' instead of 'optimizing'.

C. It will focus too much on the welfare of the 'entire organization'.

D. It may become intractable, meaning it cannot be solved with reasonable effort.

31. In the initial phase of an OR study, what is primarily meant by the statement that the OR team works in an 'advisory capacity'?

A. The team only provides data and does not suggest specific alternative courses of action.

B. The team performs technical analysis and presents recommendations, but management makes the final decision.

C. The team is responsible for both defining the problem and executing the final solution independently.

D. The team functions as a subset of the management group with full voting power on policy changes.

32. Why might an OR study avoid formulating objectives that represent the entire organization, even though doing so is theoretically ideal?

A. Mathematical models are incapable of handling more than one objective at a time.

B. The analysis might become unwieldy if it tries to account for all side effects across the whole organization.

C. Management usually prefers specific, short-term goals over long-term organizational welfare.

D. Focusing on the entire organization typically leads to the problem of suboptimization.

33. Which term describes the approach where managers seek a solution that is 'good enough' rather than mathematically optimal?

A. Suboptimizing

B. Postoptimality analysis

C. Heuristic procedure

D. Satisficing

34. In a mathematical model, what role do the variables represent?

A. The objective function that measures the overall performance.

B. Parameters that represent fixed constants provided by management.

C. Decision variables that represent quantifiable choices to be made.

D. Constraints that define the limits on what can be done.

35. What is the primary purpose of conducting a 'retrospective test' during the model validation phase?

A. To determine the absolute magnitude of the performance measure for the real-world system.

B. To see how well the model would have performed if it had been used in the past using historical data.

C. To identify which parameters are most sensitive to changes in the environment.

D. To document the mathematical steps taken so the study can be reproduced by others.

36. When an OR team encounters a problem where a standard algorithm cannot find an optimal solution in a reasonable timeframe, they might use:

A. Linear programming models

B. Model enrichment

C. Heuristic procedures

D. Sensitivity analysis

37. Sensitivity analysis is a critical part of an OR study because it helps to identify 'sensitive parameters.' According to the text, a parameter is considered sensitive if:

A. It represents a policy decision that can be easily adjusted by management.

B. It has a high correlation with the performance measure of the system.

C. Its value cannot be changed without changing the optimal solution.

D. It is difficult to estimate accurately because of 'soft' data.

38. Which phase of an OR study is often considered the most critical because it is where the benefits of the study are actually reaped?

A. Defining the problem

B. Testing the model

C. Formulating a mathematical model

D. Implementation

39. The Monsanto case study involved minimizing the cost of meeting a production target for maleic anhydride. Which component of their model was represented by the formula ?

A. The parameters

B. The constraints

C. The decision variables

D. The objective function

40. In the IBM case study, what unique approach was taken during the model validation phase to ensure the system would be successful?

A. A 'preloader program' was used to check for data errors and inconsistencies in the mainframe database.

B. Future users were incorporated into a 'user team' to serve as advisers to the OR team.

C. The OR team used a series of 100 programs to process over 15 gigabytes of data for the final run.

D. The system was tested using 4,000 pages of documentation to ensure reproducibility.

ANSWERS

1 – D, 2-C , 3- D, 4- D , 5- D, 6- A, 7- C, 8-B, 9- D, 10-C, 11- A, 12- B, 13- D, 14-D,

15-B, 16-B, 17- C, 18- D, 19-A , 20- C, 21- B, 22- B, 23- C, 24- D, 25- A, 26- D,

27- A, 28- A, 29- A, 30- D, 31- B, 32- B, 33- D, 34- C, 35- B, 36- C, 37- C, 38- D,

39- D, 40- B.